

CSA0203- C PROGRAMMING

Smart Farm Monitoring System Using C

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Introduction



Smart Farm Monitoring

Modern farms use multiple sensor nodes for monitoring.

Sensors measure:

- Soil Moisture
- Temperature
- Humidity
- Water Level

Sensor data is periodically sent to a central monitoring system.

The number of sensors can increase dynamically.

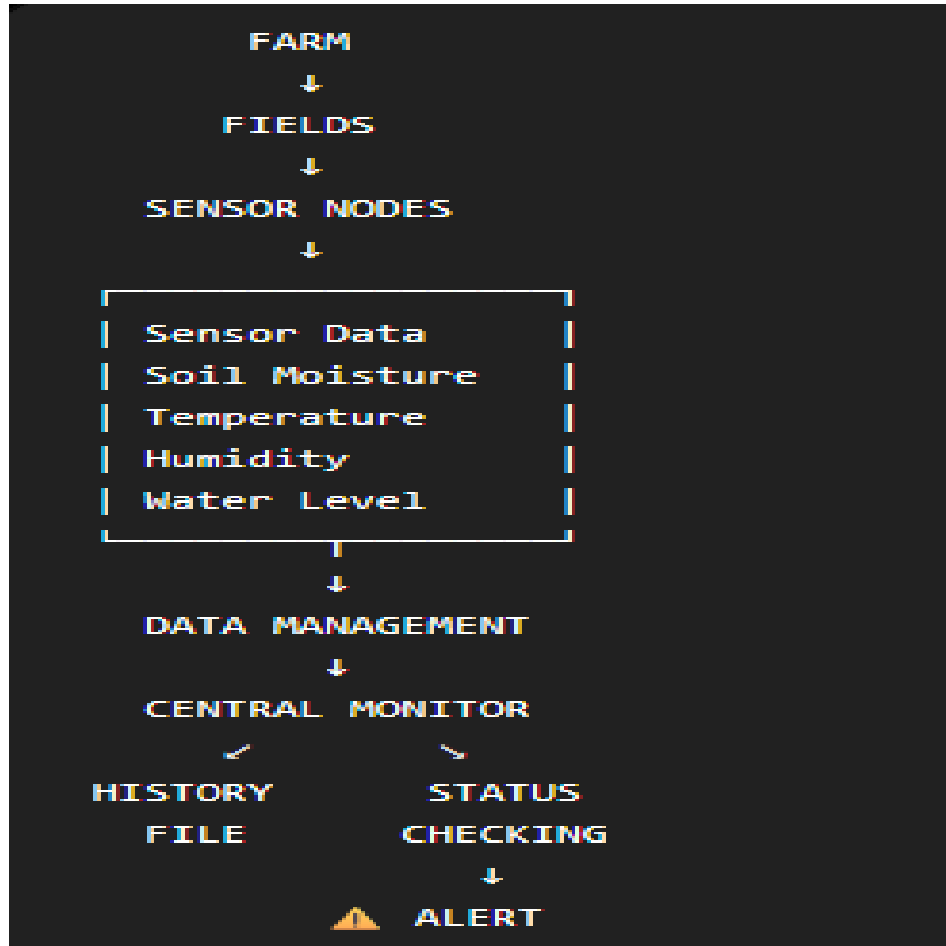
Problem Statement & Objectives

Large agricultural farms contain hundreds of sensors. Manually monitoring all sensors is difficult, especially when sensors become inactive or stop sending data.

Objectives

- Dynamically register sensor nodes.
- Manage sensor information using pointers.
- Analyse sensor readings using C functions.
- Organize farm hierarchy using recursion.
- Store historical sensor readings.
- Detect inactive sensors.

System Architecture



Key Components

- Sensor Registration
- Data Generation
- Data Management
- Historical Storage
- Sensor Status Monitoring
- Alert Generation

Functionality

- Dynamic Sensor Registration
- Sensor Data Monitoring
- Pointer-Based Data Management
- Farm Hierarchy Traversal
- Data Transmission
- Historical Data Storage
- Inactive Sensor Detection
- Automatic Alert
- Field Configuration



OUTPUT

```
main.c
265 {
285 {
301 {
318     case 5:
322     case 6:
323         free(sensors);
324
325         printf(
326             "\nSmart Farm System closed.\n"
327         );
328
329         return 0;
330
331     default:
332         printf("\nInvalid choice!\n");
333     }
334 }
335 }
```

Run

Output

```
=====
SENSOR DATA TRANSMISSION
=====
[Sensor 101] soil = 84.00
[Sensor 102] temperature = 43.00
[Sensor 103] NO DATA RECEIVED
[Sensor 104] water = 43.00

Data transmission completed.

=====
MENU
=====
1. Register Sensor
2. Display Sensors
3. Transmit Sensor Data
4. Save Historical Data
```

OUTPUT

main.c	Run	Output
<pre>265 { 285 { 301 { 318 case 5: 322 case 6: 323 free(sensors); 324 325 printf(326 "\nSmart Farm System closed.\n" 327); 328 329 return 0; 330 331 default: 332 printf("\nInvalid choice!\n"); 333 } 334 } 335 }</pre>		<pre>Enter choice: 5 ===== INACTIVE SENSOR CHECK ===== Sensor 101 : ACTIVE Sensor 102 : ACTIVE *** ALERT *** Sensor ID : 103 Sensor Type : humidity Status : INACTIVE No data received within interval. Sensor 104 : ACTIVE ===== MENU</pre>

Conclusion

- Developed a Smart Farm Monitoring System using C.
- Supports dynamic sensor registration and data management.
- Monitors soil moisture, temperature, humidity, and water level.
- Stores sensor readings for historical monitoring.
- Detects inactive sensors when data is not received within the specified interval.
- Automatically generates an alert for inactive sensors.
- The prototype demonstrates how C programming can support scalable and automated farm monitoring.

References

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THANK YOU